presence of living organisms like algae, protozoa and fungus. Most of the bacteria do not cause harm. Moreover they help in the decomposition of organic wastes naturally. Because of this, the cost of treatment also gets reduced.

5.5.2 Analysis of Sewage

Objectives

The analysis of sewage is carried out with the following objectives

(i) To ascertain various substances present in the sewage

(ii) To determine their physical, chemical and biological characteristics

(iii) To obtain information regarding the operation and maintenance of the sewage treatment works.

Types of Analysis

1. Physical analysis
2. Chemical analysis
3. Biological analysis

1. Physical analysis

This includes tests for temperature, turbidity, colour and odour. The determination of these tests are similar to that of water.

2. Chemical analysis

This includes tests for total solids, oxygen, nitrogen, pH value, chloride, fats and grease.

3. Biological analysis

Biological (or) bacteriological analysis is not considered on account of the fact that a high concentration of bacteria remains in the sewage even after the treatment.

5.6 Treatment of Sewage

Before disposing the sewage into river, stream, etc., it is essential to treat the sewage. The treatment of sewage depends on the characteristics of the wastes.

The treatment process involved the following steps.

1. Primary treatment
2. Secondary treatment
3. Tertiary treatment

Primary	Secondary	Tertiary
1. Screening	1. Filters	1. Disinfection
2. Skimming tanks	(i) Contact beds	2. Removal of Heavy Metals
3. Grit Chamber	(ii) Intermittent Sand filter	
4. Sedimentation Tank	(iii) Trickling Filter	
a. Primary Clarifier	(iv) Miscellaneous Filter	
b. Secondary Clarifier	2. Activated Sludge Process	

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